



Dual-Channel, ACTIVE AC ANALOGUE PROBE

PETRE TZV PETROV

Presented here is a low-cost, dual-channel, active AC analogue probe. It is suitable for oscilloscopes, multimeters and other analogue measurement equipment.

When experimenting with AM receivers for long, medium and short waves, users need a high-impedance analogue probe with input capacitance below 20pF. The required input impedance from the probe at low frequency is usually one mega-ohm or above.

Most of the low-cost passive probes for oscilloscopes have input capacitance well above 100pF. If you include the input capacitance of the oscilloscope, the total capacitance becomes even higher. The bandwidths of these low-cost passive probes are usually limited to 1-2MHz. These can be connected in parallel with L-C networks in AM receivers because that will change the parameters of these networks significantly.

Circuit and working

Fig. 1 shows the circuit diagram of the dual-channel, active AC analogue probe. It is built around two n-channel PN4393 JFET transistors (T1 and T2) and a few resistors and capacitors.

Most of the oscilloscopes come with two channels, so you need a dual-channel active probe with common ground and common power supply.

At low frequency, the input resistance of the probe, determined mainly by resistors R3 and R4, is around 10-mega-ohm. At high frequency, the input impedance is determined by the total input ca-

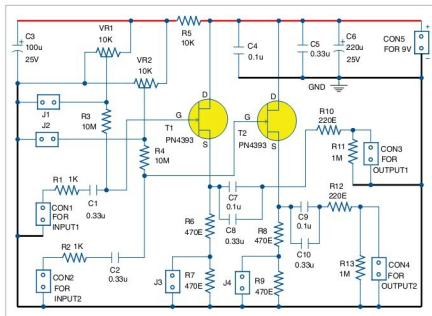


Fig. 1: Circuit diagram of dual-channel, active AC analogue probe

pacitance of the probe. With careful implementation and use of appropriate JFETs, you can obtain input capacitance below 20pF.

Transistors T1 and T2 should have low noise level and low input capacitance. So you can choose from transistors like PN4391, PN4392 and PN4393 as these have input capacitance of typically 14pF. You can also use transistors J201, J202, J203, 2N5457, 2N5458 and BF245 (with 1.5db noise figure). High-frequency JFETs in small cases are preferred but these are costly.

The voltage between the drain and the source of T1 and T2 is adjusted to around half of the power supply, using resistors R6 through R9 and jumpers J3 and J4. The circuitry using resistor R5, presets VR1 and VR2, and capacitor C3 is used to adjust the offset voltage applied

PARTS LIST

Semiconductors:

T1, T2 - PN4393 JFET

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R2 - 1-kilo-ohm
R3, R4 - 10-mega-ohm
R5 - 10-kilo-ohm
R6-R9 - 470-ohm
R10, R12 - 220-ohm
R11, R13 - 1-mega-ohm
VR1, VR2 - 10-kilo-ohm preset

Capacitors:

C1, C2, C5, C8, C10 - 0.33 μ F ceramic disk
C3 - 100 μ F, 25V electrolytic
C4, C7, C9 - 0.1 μ F ceramic disk
C6 - 220 μ F, 25V electrolytic

Miscellaneous:

CON1-CON4 - 2-pin connector
CON5 - 2-pin terminal connector
J1-J4 - Shorting jumper
- 9V DC power supply/
9V battery

to the gates of the JFETs. These components are not mandatory. If you choose not to use these, close